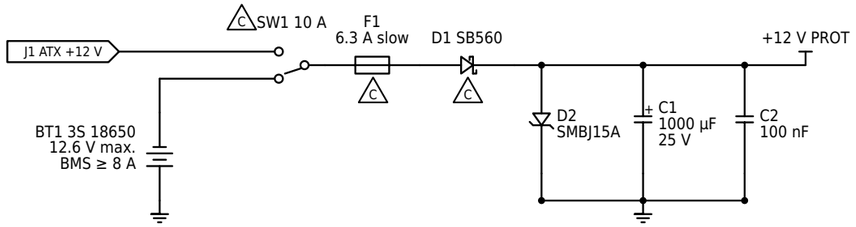
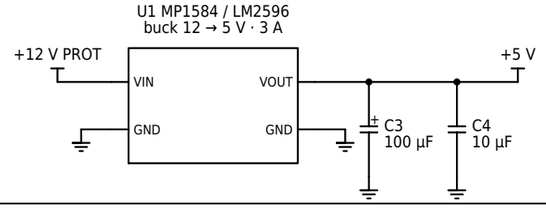
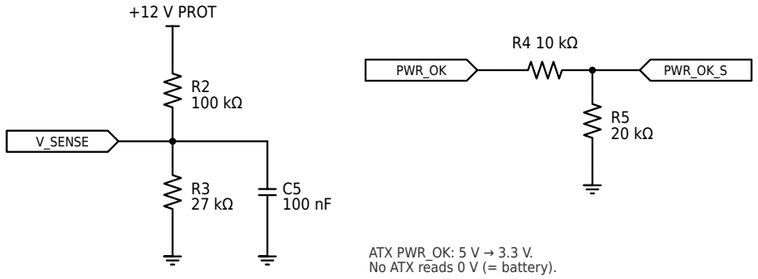
[1] INPUT · SOURCE SELECTION · PROTECTION



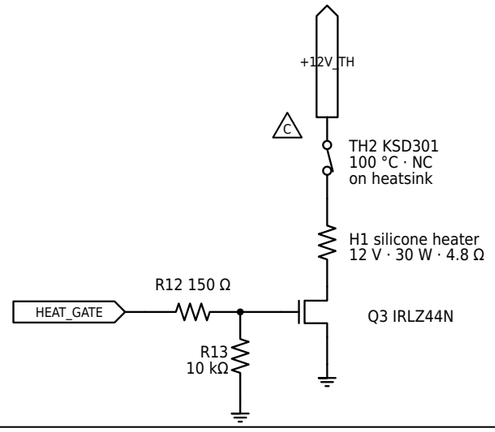
[2] 12 V → 5 V REGULATION



[5] ANALOG SENSING



[9] BASE HEATER (LOW-SIDE)



RESISTIVE load: no flyback diode. Same command as the lamps (LEDC channel 1, same timer as channel 0).

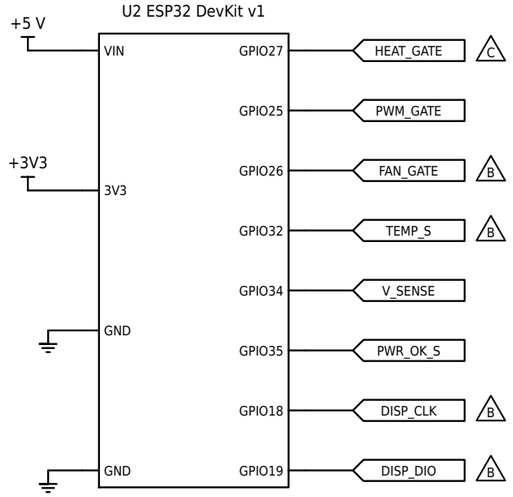
Heater bonded under a finned heatsink; insulation below protects the base (note 9).

LEGEND

General notes and power nets: sheet 2.

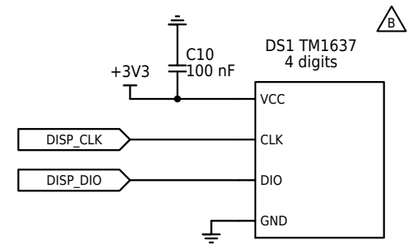
C revision change · net label (connects by name) · common ground · junction

[3] CONTROLLER



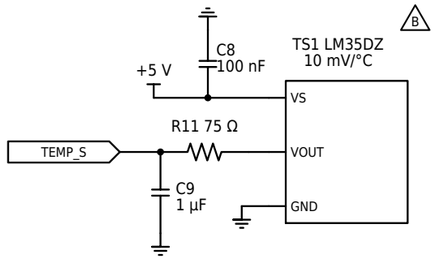
GPIO2: on-board status LED.
GPIO33 and GPIO13: reserved, not connected.

[8] TEMPERATURE DISPLAY



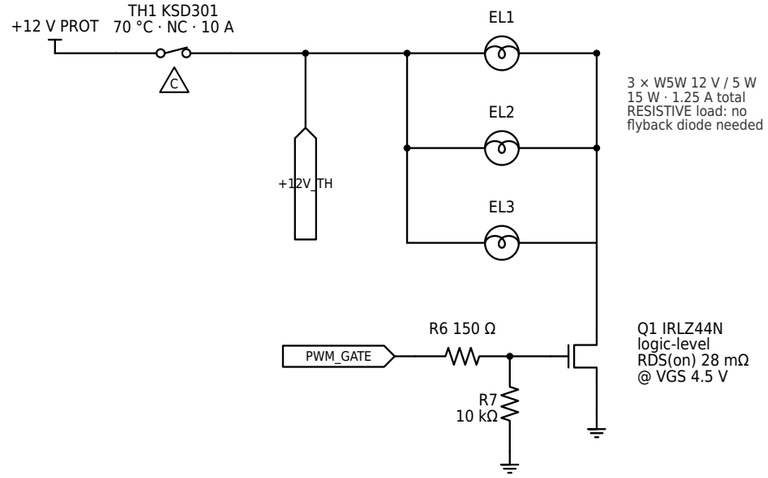
VCC at 3.3 V: the ESP32 does not tolerate 5 V on DIO (note 6).

[4] TEMPERATURE SENSOR

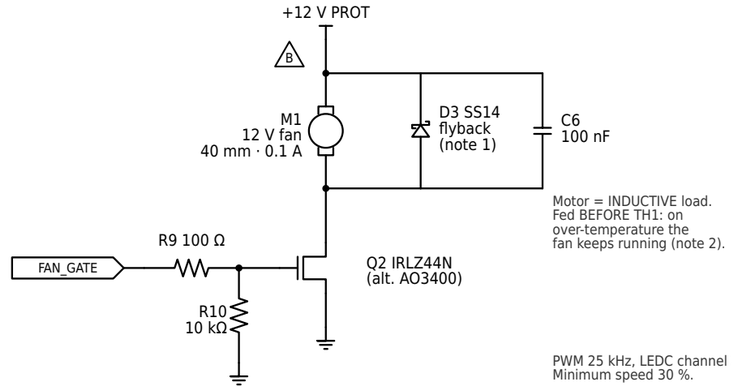


R11 + C9 filter per datasheet: the LM35 oscillates with a capacitor right at VOUT.

[6] POWER STAGE · LAMPS (LOW-SIDE)



[7] EXHAUST FAN (LOW-SIDE)



REVISIONS

REV.	DESCRIPTION	DATE	BY
A	Initial release.	10/10/2026	GM
B	Fan M1 with Q2, R9, R10, flyback diode D3 and C6. Display DS1 (TM1637) with C10. LM35DZ (TS1) replaces DS18B20, with C8 and R11/C9 filter. ATX input = J1. Removed R1, R8, LED1.	11/10/2026	GM
C	Silicone heater H1 (12 V, 30 W) under a heatsink, with Q3, R12, R13 and TH2 (KSD301 100 °C). TH1 becomes KSD301 10 A and cuts lamps and heater. F1 6.3 A, D1 SB560, SW1 10 A. GPIO27 = HEAT_GATE.	11/10/2026	GM

	NAME	INIT.	DATE
DRAWN	Gabriel Marques	GM	11/10/2026
APPROVED	Gabriel Marques	GM	11/10/2026

PROJECT PIDlampada

COMPACT THERMAL CHAMBER WITH PWM CONTROL

ELECTRICAL SCHEMATIC

UNIP · COMPUTER ENGINEERING · CAPSTONE PROJECT 2026 — SÃO JOSÉ DOS CAMPOS - SP · BRAZIL

DRAWING No.	REV.	SHEET	SCALE	SIZE	DATE
PIDL-ESQ-001	C	1 / 2	NTS	A3	11/10/2026

PINOUT · U2 ESP32 DEVKIT V1

PIN	NET	TYPE	FUNCTION	REV.
GPIO25	PWM_GATE	output	Q1 gate (lamps) · LEDC channel 0, 500 Hz, 10 bit	A
GPIO27	HEAT_GATE	output	Q3 gate (heater) · LEDC channel 1, same timer as the lamps	△C
GPIO26	FAN_GATE	output	Q2 gate (fan) · LEDC channel 2, 25 kHz, 8 bit	△B
GPIO32	TEMP_S	analog	ADC1_CH4 ← LM35 through R11/C9 filter	△B
GPIO34	V_SENSE	analog	ADC1_CH6 ← R2/R3 divider (bus voltage)	A
GPIO35	PWR_OK_S	analog	ADC1_CH7 ← R4/R5 divider (ATX PWR_OK)	A
GPIO18	DISP_CLK	output	TM1637 clock (DS1)	△B
GPIO19	DISP_DIO	bidir.	TM1637 data, open drain, pull-up on module	△B
GPIO2	–	output	On-board status LED	△B
GPIO33	SETPOINT	analog	Reserved: setpoint potentiometer (simulation only)	A
GPIO13	BTN_MODE	input	Reserved: mode button (was GPIO27)	△C
VIN	+5 V	supply	5 V input from U1	A
3V3	+3V3	supply	On-board regulator output → DS1	A
GND	GND	ground	Common ground	A

DESIGN NOTES

1. **Why a flyback diode only now.** A halogen lamp is a resistive load and needs no diode. The fan motor is inductive: when Q2 turns off the winding current needs a path and, without D3, becomes a voltage spike that can exceed the 55 V MOSFET rating. D3 is a Schottky because the PWM runs at 25 kHz.
2. **The fan does not go through TH1.** On over-temperature TH1 cuts lamps and heater and the firmware drives the fan at 100 % to remove heat. If TH1 also cut the fan, the chamber would only cool by convection.
3. **Split-range control.** A single PI with output from −1 to 1: positive drives the lamps, negative drives the fan, never both. The fan has a 30 % minimum speed and on/off hysteresis.
4. **LEDC channels.** On the ESP32 channels come in pairs sharing one timer (0-1, 2-3...). The fan uses channel 2 so its 25 kHz does not change the lamp frequency on channel 0.
5. **LM35 at 5 V.** The LM35 needs 4 to 30 V. Its output, 0.25 to 0.60 V over the useful range, goes to the ADC with no divider. R11 and C9 follow the datasheet circuit.
6. **TM1637 at 3.3 V.** The module pull-ups go to VCC. With the module at 5 V the ESP32 DIO pin would see more than its limit.
7. **Kept from revision A.** D1 against reverse polarity (SB560 since C, for the higher current), SMBJ15A TVS against hot-plug transients, slow-blow fuse for inrush, R7, R10 and R13 holding the gates at 0 V during boot.
8. **Voltage compensation.** On battery the bus drops from 12.6 to ~9 V. The firmware scales the lamp and heater duty by $(12/V)^2$ and the fan duty by 12/V, reading V on V_SENSE.
9. **Silicone heater in the base.** H1 is bonded under a finned aluminium heatsink: it spreads the heat evenly and exchanges it with the air over an area several times larger than the pad. Ceramic-fibre insulation below protects the acrylic base. It gets the same command as the lamps, so their brightness still shows the control signal; the feedforward uses the combined gain (30 + 60 °C).
10. **Two thermal cut-offs.** TH1 (KSD301 70 °C, in the air) cuts lamps and heater. TH2 (KSD301 100 °C, on the heatsink) cuts only the heater if the heatsink overheats with a loose sensor or still air. The firmware also cuts everything above 68 °C. The 60 °C setpoint ceiling comes from the acrylic, not from the power.

GENERAL NOTES

1. Blocks [1] to [9] on sheet 1, connected to each other by net labels.
2. Triangle with a letter: part or connection added or changed in that revision (B or C).
3. Net labels with the same name are connected.
4. All ground symbols are the same node: pack, ATX, U1 and U2 GND.
5. Resistors 1/4 W 5 %. Ceramic capacitors X7R 50 V. Unless noted.
6. Power loop (+12 V PROT, TH1, EL1-EL3, H1, Q1, Q3): conductors rated 5 A minimum.

POWER NETS

NET	SOURCE	LOADS	CURRENT
+12 V PROT	J1 or BT1, via SW1, F1, D1	TH1, M1, U1, R2/R3	≈ 4 A max.
+12V_TH	TH1 (KSD301 70 °C)	EL1-EL3, TH2 + H1	≈ 3.75 A max.
+5 V	U1 (buck)	U2 (VIN), TS1	≈ 0.3 A peak (Wi-Fi)
+3V3	U2 on-board regulator	DS1	≤ 30 mA

BILL OF MATERIALS

ITEM	REF.	QTY	DESCRIPTION	RATING	REV.
1	**BT1**	1	3S 18650 battery pack with BMS	12.6 V max. · BMS ≥ 8 A	A
2	**J1**	1	ATX input connector (4-pin Molex)	+12 V / GND	△B
3	**SW1**	1	SPDT selector switch	10 A · ATX / battery	△C
4	**F1**	1	Slow-blow fuse	6.3 A	△C
5	**D1**	1	SB560 Schottky diode	5 A · 60 V	△C
6	**D2**	1	SMBJ15A TVS suppressor	15 V · 600 W	A
7	**C1**	1	Electrolytic capacitor	1000 µF · 25 V	A
8	**C2, C5**	2	Ceramic capacitor	100 nF · 50 V	A
9	**U1**	1	MP1584 / LM2596 buck module	12 → 5 V · 3 A	A
10	**C3**	1	Electrolytic capacitor	100 µF · 16 V	A
11	**C4**	1	Ceramic capacitor	10 µF · 16 V	A
12	**U2**	1	ESP32 DevKit v1 board	30-pin	A
13	**R2**	1	Resistor	100 kΩ	A
14	**R3**	1	Resistor	27 kΩ	A
15	**R4, R7**	2	Resistor	10 kΩ	A
16	**R5**	1	Resistor	20 kΩ	A
17	**R6**	1	Resistor	150 Ω	A
18	**TH1**	1	KSD301 bimetal thermostat (in the air)	70 °C · NC · 10 A	△C
19	**EL1-EL3**	3	W5W halogen lamp with T10 socket	12 V · 5 W	A
20	**Q1**	1	IRLZ44N N-channel MOSFET	logic-level · TO-220	A
21	**TS1**	1	LM35DZ temperature sensor	10 mV/°C · TO-92	△B
22	**C6, C8, C10**	3	Ceramic capacitor	100 nF · 50 V	△B
23	**C9**	1	Ceramic capacitor	1 µF · 16 V	△B
24	**R11**	1	Resistor	75 Ω	△B
25	**M1**	1	40 × 40 × 10 mm fan, 2-wire	12 V · 0.1 A	△B
26	**Q2**	1	IRLZ44N N-channel MOSFET (alt. AO3400)	logic-level	△B
27	**D3**	1	SS14 Schottky diode (alt. 1N5819)	1 A · 40 V	△B
28	**R9**	1	Resistor	100 Ω	△B
29	**R10**	1	Resistor	10 kΩ	△B
30	**DS1**	1	TM1637 display module	4 digits with point	△B
31	**H1**	1	100 × 100 mm silicone heater pad	12 V · 30 W · 4.8 Ω	△C
32	**—**	1	100 × 100 mm finned aluminium heatsink	on top of H1	△C
33	**—**	1	Ceramic-fibre insulation (2 mm paper)	under H1	△C
34	**TH2**	1	KSD301 bimetal thermostat (on heatsink)	100 °C · NC · 10 A	△C
35	**Q3**	1	IRLZ44N N-channel MOSFET	logic-level · TO-220	△C
36	**R12**	1	Resistor	150 Ω	△C
37	**R13**	1	Resistor	10 kΩ	△C

Removed in revision B: R1 (1-Wire pull-up), R8 and LED1 (external status LED) and the DS18B20 sensor. Changed in C: F1 (was 3 A), D1 (was SS34), SW1 (was 3 A), TH1 (was KSD9700).

	NAME	INIT.	DATE
DRAWN	Gabriel Marques	GM	11/10/2026
APPROVED	Gabriel Marques	GM	11/10/2026

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ELECTRICAL SCHEMATIC · PINOUT · DESIGN NOTES · BILL OF MATERIALS

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DRAWING No.	REV.	SHEET	SCALE	SIZE	DATE
PIDL-ESQ-001	**C**	2 / 2	NTS	A3	11/10/2026